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Biofilm-forming ability, antibiotic resistance and phylogeny of *Escherichia coli* isolated from extra intestinal infections of humans, dogs, and chickens

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ABSTRACT

Escherichia coli (*E. coli*) causes various infections in humans and animals. The biofilm-forming ability of *E. coli* has increased antimicrobial resistance and capacity to cause recurrent and chronic infections. This study determined the biofilm-forming ability of *E. coli* isolated from extraintestinal infections of humans, chickens, and dogs in relation to the phylogroup, type of infection, and antibiotic resistance. Isolates from chickens showed significantly higher biofilm-forming ability compared to those causing urinary tract infections in humans ($p = 0.0001$). Further, isolates belonging to phylogroup B1 displayed a higher likelihood to form biofilms. Resistance to ciprofloxacin and trimethoprim-sulfamethoxazole was positively correlated with biofilm-forming ability. Harboring plasmid-mediated quinolone resistance gene, *qnrS* was also positively correlated with biofilm formation. This study provides insight into factors such as phylogroup and the type of infections that could enhance biofilm formation, as well as genotypic and phenotypic antibiotic resistance that could correlate with the ability to form biofilms.

1. Introduction

Escherichia coli is a ubiquitous organism with a vast pathogenic potential. Based on the disease-causing ability, *E. coli* is broadly categorized as nonpathogenic or commensal *E. coli*, intestinal pathogenic *E. coli* (InPEC), and extraintestinal pathogenic *E. coli* (ExPEC) [1]. Urinary tract infections (UTIs) in humans and animals caused by uropathogenic *E. coli* (UPEC), and colibacillosis in poultry caused by avian pathogenic *E. coli* (APEC) are extensively studied due to their high incidence rates and economic impact [2,3].

UPEC is responsible for urinary tract infections of 80–90% cases of over 150 million humans reported annually [4]. APEC is associated with severe economic losses to the commercial chicken industry through high mortality rates, decreased production, carcass condemnation, and treatment costs [5]. Studies have shown that APEC and UPEC often belong to phylogroups B2 and D [3]. Pathogenicity of APEC and UPEC is

linked to various virulence factors (i.e. toxins, adhesions, lipopolysaccharides, polysaccharide capsules, proteases, siderophores, serum survival) that aid in overcoming host defences, allowing colonization or invasion in various tissues [6].

Until recently, diseases caused by ExPEC were effectively treated with first-line antibiotics. However, ExPEC strains have now become resistant to these antibiotics, including trimethoprim-sulfamethoxazole, extended-spectrum cephalosporins (ESCs), and fluoroquinolones (FQs) [7,8]. Misuse of antibiotics, inherent or acquired genetic changes such as mutations of target genes, and exchange of resistance genes through mobile genetic elements are well-known causes of antimicrobial resistance.

Recently, biofilms have also been identified as a key mediator for development of AMR [9]. Biofilms prevent antimicrobial agents, antibodies, and immune cells from reaching the organisms on living tissues, while they shield microbial communities from the actions of

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disinfectants, sanitisers, and thermal treatments on abiotic surfaces [10, 11]. Biofilm formation is particularly common in indwelling medical devices such as urinary catheters, which are a major source of hospital-acquired UTI [9,12]. Biofilms play a crucial role in harbouring APEC in water feeding systems in poultry that include devices such as waterers, drinkers, and water distribution pipes [13,14].

Biofilm-forming ExPEC strains often cause chronic and recurrent infections due to their enhanced ability to colonise and persist in tissues in contrast to planktonic cells. They are also more resistant to immune responses and antimicrobial therapy. In addition to forming extracellular biofilms, UPEC can form biofilm-like intracellular bacterial communities (IBCs) with the help of type I fimbriae [15,16]. Biofilm bacteria have a higher likelihood of acquiring transmissible genetic elements. Since the antibiotic resistance of APEC is mainly mediated through large transmissible plasmids, the biofilm-forming *E. coli* is reported to have increased potential of multidrug resistance [17,18].

Several studies have reported significant variations among ExPEC strains in their ability to form biofilms [19,20]. Knowledge on the variations in biofilm-forming ability of *E. coli* with regard to infection types, phylogroups, and host species is important for formulating treatment, improving patient outcomes, and implementing appropriate infection control strategies.

The core objective of the present study was to identify the biofilm-forming ability of *E. coli* that causes extraintestinal infections such as UTI and uterine infections in humans and dogs, as well as colibacillosis in chickens. Furthermore, the correlation between biofilm-forming ability and disease category, phylogroup, and antimicrobial resistance was evaluated to identify the common traits of biofilm-forming ExPEC.

2. Materials and methods

2.1. Bacterial isolates

Coliform isolates of cultured urine or uterine content of humans and dogs were obtained from the microbiology laboratories at the Medical Teaching Hospital and the Veterinary Teaching Hospital (Peradeniya, Sri Lanka) during 2020 and 2021. In addition, coliforms isolated from the liver, bone marrow, and heart blood of diseased chickens with clinical signs and post-mortem lesions suggestive of colibacillosis were obtained from the Veterinary Research Institute (Peradeniya, Sri Lanka). All isolates were sub-cultured on *E. coli* chromogenic medium and confirmed by a PCR assay as described in Clermont et al. (2009) [21]. The 427 bp fragment of *trpA* gene of *E. coli* was amplified with primers *trpA*. F (5'-GCTACGAATCTCTGTTTGCC-3') and *trpA*2. R (5'-GCAACGCGGCCTGGCGGAAG-3'). Each reaction was carried out in a 20 µl mixture containing 5x buffer, 1.5 mM MgCl₂, 12 pmol of each of the *trpA* primers, 2 mM each dNTP, and 1 U of Taq DNA polymerase. The PCR amplicons were separated by electrophoresis on 1.5% agarose gel. *E. coli* isolates from humans (n = 73) and dogs (n = 21) with UTI or endometritis, and commercial chickens with colibacillosis (n = 38) were included in the study. The total number of isolates included was 132.

2.2. Antimicrobial susceptibility testing (AST)

Antimicrobial susceptibility of the *E. coli* isolates to ampicillin (AMP:10 µg), amoxicillin-clavulanate (AMC: 20–10 µg), cefuroxime (CXM: 30 µg), ceftriaxone (CRO:30 µg), cefotaxime (CTX: 5 µg), ceftazidime (CAZ: 30 µg), cefepime (FEP: 30 µg), cefpodoxime (CPD:10 µg) ciprofloxacin (CIP: 5 µg), trimethoprim-sulfamethoxazole (SXT: 1.25/23.75 µg), imipenem (IMP:10 µg), amikacin (AK: 30 µg), and gentamicin (CN: 10 µg) (Oxoid, UK) were determined by disk diffusion method following the guidelines of European Committee on Antimicrobial Susceptibility Testing (EUCAST Version 7, January 2019) (<http://www.eucast.org>). Isolates showing resistance to three or more antimicrobial classes were identified as multidrug resistant (MDR) strains.

2.3. Identification of the phylogenetic group of *E. coli*

The quadruplex PCR described by Clermont et al. [22] was used to assign *E. coli* strains into the phylogroups (A, B1, B2, C, D, E, F, and cryptic clade). PCR mixtures (20 µl) consisted of 4 µl of 5x buffer, 2 µl of 25 mM MgCl₂, 20 pmol of each primer, each dNTP of 5 µM, 2.5 U of Taq polymerase (Himedia HigenoMB®) and 1 µl of genomic DNA. PCR was performed in a thermal cycler (Bioer®, China) in 0.2 mL PCR tubes under the following conditions: denaturation for 4 min at 94 °C, 30 cycles of 30 s at 94 °C, 30 s at 55 °C and 30 s at 72 °C, and final extension of 7 min at 72 °C.

2.4. Identification of plasmid-mediated quinolone resistance encoding genes

A multiplex PCR assay was used to detect three plasmid-mediated quinolone resistance encoding genes: *qnrA*, *qnrB*, and *qnrS*. The multiplex PCR was performed in a 50 µl reaction mixture containing 10x buffer with 25 mM MgCl₂, 250 µM each of deoxynucleotide triphosphate, 10 pmol of each of the six primers (supplementary Table 1), and 1 U of Taq DNA polymerase.

Amplification was carried out with the following thermal cyclic conditions: Initial denaturation for 4 min at 94 °C and 35 cycles of amplification consisting of 1 min at 94 °C, 1 min at 54 °C and 1 min at 72 °C and 10 min at 72 °C for the final extension. DNA fragments were analysed by gel electrophoresis on 2% agarose gel. Fragments were stained with Promega Diamond® nucleic acid dye and visualised using a blue light transilluminator (Clear Scientific® safe, mini view).

2.5. Identification of β -lactam resistance genes in ExPEC isolates

Three separate PCR assays were run to detect the three β -lactam resistance encoding genes, namely *bla*_{TEM}, *bla*_{SHV}, and *bla*_{CTX-M} in ExPEC isolates using the primer sequences listed in the supplementary Table 1. The PCR were carried out in a 25 µl reaction mix containing 5 µl 5x buffer, 1 µl 25 mM MgCl₂, 1 µl of each primer, 0.2 µl dNTP, 0.5 U of Taq DNA polymerase (Himedia HigenoMB®) and 200 ng of genomic DNA (Primers used for each PCR are given in Supplementary Table 1).

2.6. Biofilm assay

The ability of the *E. coli* isolates to form biofilms was assessed using the crystal violet colorimetric assay described previously [24] with slight modifications. *E. coli* ATCC 25922 was used as a positive control. *E. coli* isolates were cultured overnight in Trypticase soy broth (TSB), and 10 µl of each overnight culture was added to 1 mL TSB broth in glass tubes and turbidity was adjusted to 0.5 McFarland Standard. A volume of 200 µl was transferred into flat bottom microtiter plates in triplicate and incubated at 37 °C for 24 h without shaking. The microtiter plates were washed three times with sterile distilled water to remove all non-adherent bacteria and the attached bacteria were fixed by adding 200 µl of methanol for 15 min. After the removal of excess liquid, wells were washed three times with sterile distilled water and stained for 5 min with 200 µl of 1% crystal violet at room temperature. After staining with crystal violet, the extra stain was rinsed off by placing the microtiter plate under running tap water and then air-dried at room temperature. The crystal violet was solubilized with 160 µl of 33% glacial acetic acid and the absorbance was measured with an automated microplate reader at optical density of 560 (OD₅₆₀) nm measurement. The ODs for each test strain were averaged and normalized against the negative control by subtracting the average OD of the negative control from the average OD of the test strain. All strains were classified into four categories according to the normalized OD: negative (OD₅₆₀ < 0.01), weak (OD₅₆₀: 0.02–0.2), moderate (OD₅₆₀: 0.2–0.4), or strong biofilm producers (OD₅₆₀ > 0.4).

Table 1
Biofilm-forming ability of the *E. coli* isolates in relation to the phylogroups.

Phylogroup	Negative	Weak	Moderate	Strong	Total
A	1 (10%)	2 (20%)	4 (40%)	3 (30%)	10
B1	2 (11.7%)	3 (17.6%)	2 (11.7%)	10 (58.8%)	17
B2	19 (25.3%)	21 (28%)	17 (22.7%)	18 (24%)	75
D	2 (28.6%)	-	4 (57.1%)	1 (14.3%)	7
E	5 (21.7%)	5 (21.7%)	4 (17.4%)	9 (39.1%)	23
Total	29 (22%)	31 (23.5%)	31 (23.5%)	41 (31%)	132

2.7. Statistical analysis

Statistical testing was performed using the R software. The Kruskal-Wallis test was used to determine significant differences in biofilm formation between phylogroups. The Mann-Whitney U test was used for group comparisons. Correlations between biofilm formation and antibiotic resistance or resistance encoding genes were analysed using the "corrplot" package in R.

3. Results

3.1. Semi-quantitative classification of biofilm using the crystal violet colorimetric assay

When the biofilm-forming ability of the *E. coli* isolates was assessed by the crystal violet assay, a diverse range of optical density (OD₅₆₀) values (0.07 to 0.73) were obtained. Based on the criteria described

previously, isolates were categorized under four groups and accordingly, 41 (31%) isolates were strong biofilm producers, 31 (23.5%) were moderate, and 31 (23.5%) were weak biofilm producers whilst 29 (22%) were unable to produce biofilm.

3.2. Association between phylogroup and biofilm-forming ability of *E. coli*

The PCR assay classified 75 isolates to phylogroup B2, 23 isolates to phylogroup E, 17 isolates to phylogroup B1, 10 isolates to phylogroup A, and seven isolates to phylogroup D. The average optical densities of the crystal violet assay for the isolates belonged to phylogroups A, B1, B2, D, and E were 0.368 ± 0.162 , 0.389 ± 0.139 , 0.316 ± 0.149 , 0.294 ± 0.138 , 0.312 ± 0.14 respectively. With respect to biofilm-forming ability relative to the phylogroup, over 70% of the isolates in the phylogroups A (70%), B1 (70.5%), and D (71.4%) were strong or moderate biofilm producers whilst only 56% isolates in the phylogroup B2 and E showing that ability (Table 1). Further, 58.8% of the isolates of phylogroup B1 produced strong biofilms.

When the biofilm-forming ability of the isolates of the different phylogroups were compared to each other (Fig. 1), *E. coli* of phylogroup B1 formed biofilm more frequently than others and showed a trend to be significant more frequently than phylogroup B2 strains (Kruskal-Wallis test, $p = 0.025$).

Of the 132 isolates used in the study, 83 were isolated from urine samples of humans or dogs with UTI, 11 were from humans or dogs with endometritis and 38 were from commercial layer and broiler chickens with colibacillosis. Biofilm-forming ability of those isolates relative to the type of infection is summarized in Table 2. Over 97% of isolates

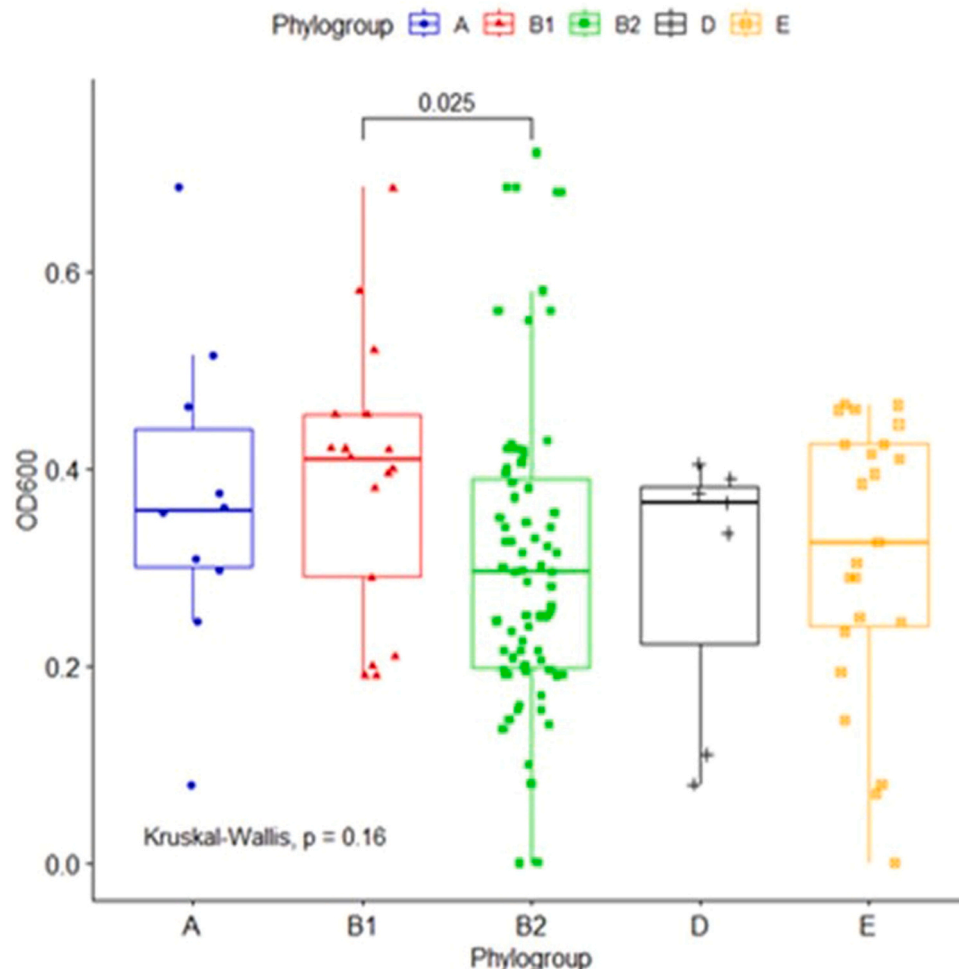


Fig. 1. Biofilm-forming ability of *E. coli* in relation to the phylogroup.

Table 2

Antimicrobial resistance of ExPEC isolates according to the bio film forming ability.

Biofilm-forming <i>E. coli</i> (%)				
Antibiotic	Negative (N = 29)	Weak (n = 31)	Moderate (n = 31)	Strong (n = 41)
Ampicillin	89.5%	81.2%	96.7%	90.2%
Amoxicillin-clavulanate	55.1%	50.0%	77.4%	78.0%
Trimethoprim-sulfamethoxazole	34.4%	53.1%	77.4%	87.8%
Ciprofloxacin	34.4%	59.3%	80.6%	78.0%
Cefotaxime	48.2%	46.9%	38.7%	29.2%
Ceftriaxone	31.0%	37.5%	32.2%	36.5%
Cefepime	34.4%	40.6%	32.2%	24.3%
Ceftazidime	41.3%	43.7%	45.1%	46.3%
Cefpodoxime	34.4%	46.8%	70.9%	48.7%
Gentamicin	20.7%	25.0%	48.3%	36.5%
Amikacin	13.8%	15.6%	22.5%	7.3%
Imipenem	6.9%	9.3%	32.5%	4.8%

obtained from commercial chickens were strong/moderate biofilm producers whereas only 45.5% endometritis isolates and 35.4% UTI isolates showed that potential.

With respect to biofilm-forming ability of *E. coli* relative to different infection categories, *E. coli* isolated from colibacillosis formed biofilm with significantly higher frequency ($p < 0.0001$) than the UTI isolates (Fig. 2).

The analysis of antimicrobial resistance among biofilm-forming *E. coli* revealed resistance rates over 89% for ampicillin, and between

64–66% for amoxicillin-clavulanate, trimethoprim-sulfamethoxazole, and ciprofloxacin. Resistance rates of 32.5–50% were observed with respect to extended spectrum beta-lactamases such as cefepime, ceftazidime, and cefpodoxime.

When the isolates were categorised into four groups based on biofilm-forming ability, strong or moderate biofilm-producing isolates displayed over 77% resistance to clavulanate- amoxicillin, trimethoprim-sulfamethoxazole, and ciprofloxacin. No significant variations in the frequency of resistance to the other nine antibiotics were observed between the groups. A high percentage of *E. coli* with biofilm-forming ability were found to be multidrug-resistant (i.e. strong: 84.1%; moderate: 85.7%; weak: 65.7%), while only 34.4% of non-biofilm producers were MDR.

Table 3 summarizes the antimicrobial resistance, phylogroup and biofilm-forming ability of ExPEC under respective host. The ExPEC isolates of human and dogs mainly belonged to phylogroup B2 whereas the chicken isolates were mostly in the phylogroup B1 and E. As mentioned previously, over 97% of *E. coli* isolates from chickens have moderate or high biofilm-forming ability while only 57% of dogs and 30% of humans isolates were showing moderate or high capacity to form biofilms.

Of the three *qnr* genes tested, *qnrA*, *qnrB* and *qnrS* were detected in four (UTI/endometritis=3, colibacillosis=1), 28 (UTI/endometritis=19, colibacillosis=9) and 47 (UTI/endometritis=24, colibacillosis=23) isolates respectively. The genes *bla_{TEM}*, *bla_{CTX-M}* and *bla_{SHV}* were detected in 110 (UTI/endometritis: 80, colibacillosis= 30), 58 (UTI/endometritis: 42, colibacillosis: 16) and 22 isolates (UTI/endometritis: 17, colibacillosis: 5) respectively. Fig. 3 illustrates the correlation analysis between phenotypic and genotypic antibiotic resistance and biofilm-

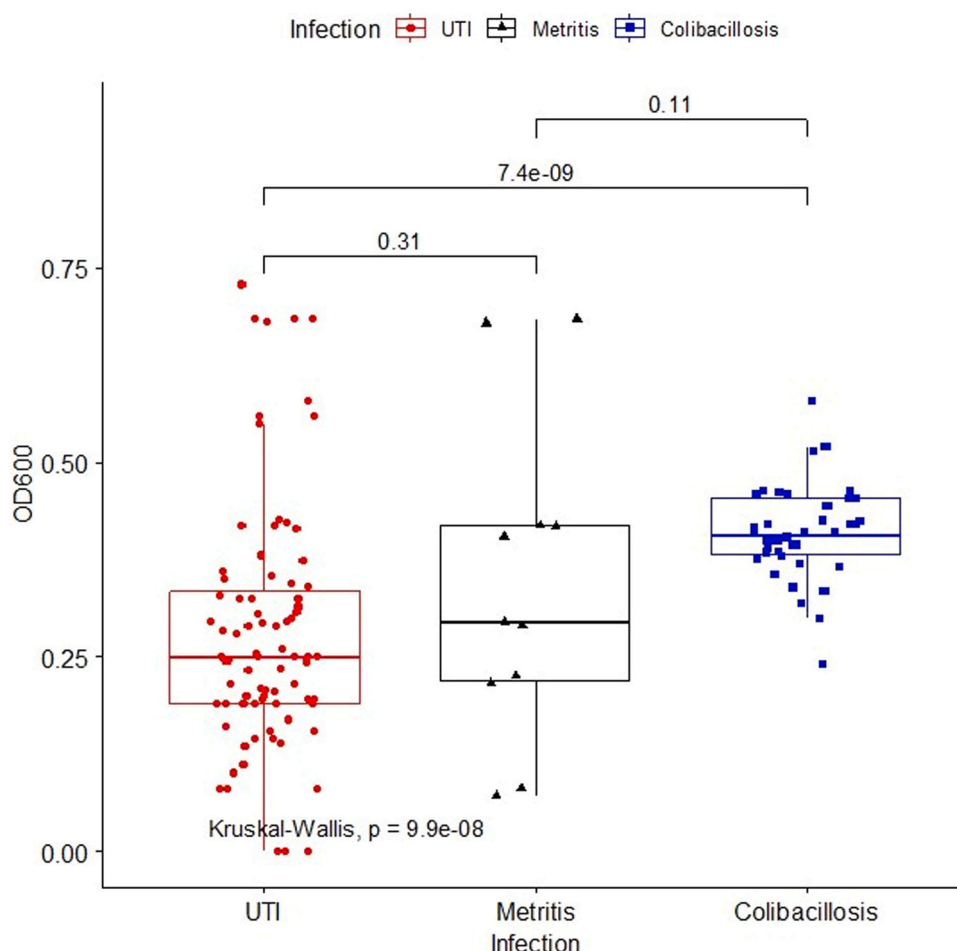


Fig. 2. Biofilm-forming ability of *E. coli* relative to the type of infection.

Table 3Phylogroup, antimicrobial resistance and biofilm forming ability of extra intestinal pathogenic *E. coli* isolated from humans, chickens and dogs.

Trait	Humans	Chickens	Dogs
Phylogroup			
A	4 (5.5%)	4 (10.5%)	2 (9.5%)
B1	2 (2.7%)	11 (28.9%)	4 (19.1%)
B2	55 (75.3%)	7 (18.4%)	12 (57.1%)
D/E	2 (2.8%)	5 (13.1%)	1 (4.8%)
E	10 (13.7%)	11 (28.9%)	2 (9.5%)
Antibiotic resistance			
Ampicillin	61 (83.5%)	36 (94.7%)	21 (100%)
Amoxicillin-clavulanate	33 (45.2%)	38 (100%)	16 (76.2%)
Trimethoprim-sulfamethoxazole	37 (50.6%)	35 (92.1%)	14 (66.6%)
Ciprofloxacin	39 (53.4%)	33 (86.8%)	12 (57.1%)
Cefotaxime	38 (52%)	2 (5.3%)	13 (61.1%)
Ceftriaxone	31 (42.4%)	2 (5.3%)	13 (61.9%)
Cefepime	30 (41.1%)	3 (7.9%)	10 (47.6%)
Ceftazidime	32 (43.8%)	15 (39.5%)	11 (52.3%)
Cefpodoxime	34 (46.5%)	18 (47.4%)	14 (66.6%)
Gentamicin	22 (30.1%)	17 (44.7%)	5 (23.8%)
Amikacin	11 (15.1%)	8 (21.1%)	0
Imipenem	8 (10.9%)	0	0
MDR	44 (60.3%)	34 (89.5%)	15 (71.4%)
Biofilm forming ability			
none	25 (34.2%)	-	4 (19.1%)
weak	26 (35.6%)	1 (2.6%)	4 (19.1%)
moderate	11 (15.1%)	14 (36.8%)	6 (28.5%)
strong	11 (15.1%)	23 (60.5%)	7 (33.3%)

forming ability of *E. coli*. Among the 12 antibiotics tested, ciprofloxacin ($r = 0.33$; $p = 0.0001$) and trimethoprim-sulfamethoxazole ($r = 0.34$; $p = 0$) exhibited a significant positive correlation with biofilm-forming ability of *E. coli*. Furthermore, a positive correlation was observed between biofilm-forming ability and multidrug resistance of the organisms ($r = 0.42$; $p = 0$). As shown in the figure, biofilm forming ability and presence of *qnrS* gene also showed a positive correlation $P = 0.001$ and $r = 0.28$. No significant positive correlation was detected in biofilm forming ability and the presence of beta lactam resistance encoding genes.

4. Discussion

Bacterial biofilms have emerged as a significant medical concern, as more than 50% of microbial infections lead to the formation of biofilms, which greatly enhance resistance to both natural defense mechanisms and antibiotics [23,24]. *E. coli* isolates used in this study also showed a considerable high potential to form biofilms, with 54% of the isolates producing moderate to strong biofilms. This finding is consistent with Rijavec et al. [25] who reported a 53% of urinary *E. coli* isolates as forming biofilms. However, other studies have reported lower [26] or higher frequencies of biofilm-forming ability in *E. coli* isolated from extraintestinal infections [27,28]. Discrepancies among studies could be attributed to variations in culture media, methods used to assess biofilm formation, and differences in the geographic and demographic characteristics of the isolates [29,30]. In our study, we used TSB as the culture medium, but it is worth noting that nutrient-depleted media have been shown to enhance biofilm formation in organisms [30,31]. Additionally, the ability to form a biofilm in one type of medium does not necessarily

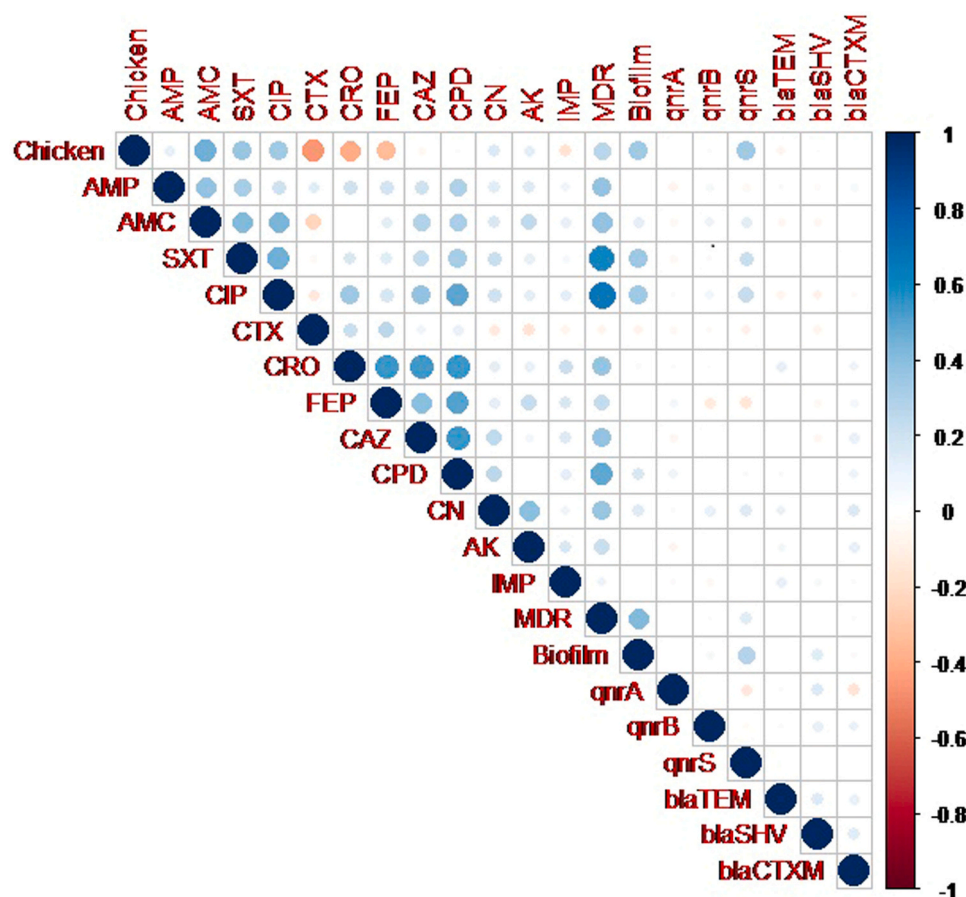


Fig. 3. Correlation matrix of phenotypic and genotypic antimicrobial resistance and biofilm-forming ability of *E. coli* isolated from extra intestinal infections of humans, dogs, and chickens. Blue circles indicate significant positive correlation and red show significant negative correlation. The size and strength of colour represent the numerical value of the Phi correlation coefficient.

indicate the same capability in another [19].

The isolates that comprised the majority of the non-biofilm producers in our study were isolated from urinary tract infections in humans. Hancock et al. [32] reported that the UPEC strains used in their study exhibited 10-fold less ability to form biofilms compared to *E. coli* isolated from asymptomatic bacteria. This suggests that biofilm formation may be a colonization strategy of commensal *E. coli* rather than UPEC. Tapiainen et al. [33] evaluated the biofilm-forming ability of *E. coli* isolated from recurrent cystitis and recurrent pyelonephritis and found poor biofilm formation in the former condition. Other studies have also shown that UPEC typically forms weaker biofilms than other *E. coli* pathotypes and forms intracellular bacterial communities more frequently [34,35].

In contrast, the majority of *E. coli* isolated from infections in chickens in our study demonstrate a high ability to form biofilms. In explaining high biofilm forming ability in avian *E. coli*, several studies have shown that some antibiotics, including fluoroquinolones when present at sub-therapeutic concentrations, can significantly induce biofilm formation in various bacterial species [36,37].

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.cimid.2023.102123](https://doi.org/10.1016/j.cimid.2023.102123).

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